

technologies are not implemented overnight. It takes time for companies to figure out how to use them. But three decades from now, when our machines are 1,000,000 times more capable than they are today, they will probably be cheaper, faster, and better at almost everything that humans can do for money.

Some people reply that we will endlessly create jobs which machines are unable to do, either because they are not creative, or because they are not conscious, and therefore have no empathy. These are likely to prove vain hopes: AI systems are creative already, and they can decode human emotions very effectively. They can also provide the same appearance of empathy that humans often do at work.

If technological unemployment is coming, we must transition to a very different type of economy. I call the transition the “economic singularity”, using a term from physics which was first applied to human affairs by computer pioneer John von Neumann. Back in the 1950s, he said we were “approaching some essential singularity in the history of the race beyond which human affairs, as we know them, could not continue.” How should policy makers respond to the challenges presented by the economic singularity? Let’s look at a few options.

SHOULD WE TRY TO PROTECT JOBS?

In America, some truckers’ groups want to ban the introduction of self-driving lorries. Not unreasonably, they argue that the technology will put millions of them out of work, with serious knock-on effects for their families and their communities. This could be an effective strategy if they were not in a competitive market: train drivers unions in the UK have prevented efficiency measures for years. But unlike trains, trucking is a competitive market. And even if the federal government were to wave a magic wand and give the truckers their wish, the US economy would become less competitive internationally. Protecting jobs stifles competition, kills innovation, and makes everyone poorer.



SHOULD WE TAX ROBOTS?

Taxing robots in order to fend off or mitigate technological unemployment would also stifle both innovation and competitiveness. It also suffers from conceptual difficulties. Machines will rarely replace humans on a one-for-one basis, and most of the “robots” automating jobs in the coming years will be software, not androids. Humans will disappear from many call centres, for instance, and be replaced by AI systems running on banks of servers in big buildings with powerful air conditioning. Does the government tax one entity – namely the AI system – or each individual server? Or does it estimate the number of humans who have been laid off by the system and calculate the tax based on that?

If we are going to tax automating machines, do we start with tractors and weaving machines? Or do we just tax new instances of replacement? Imagine

two firms making the same product. One has been going for a few years, and recently replaced half its staff with machines. The other is a startup, and used machines from the outset. The former would be hit by a tax that the latter would escape. Not only is that patently unjust, it would simply mean the former would close, and the tax base would disappear anyway.

SHOULD WE INTRODUCE A UNIVERSAL BASIC INCOME?

There are a number of problems with introducing UBI in the types of economy that exist today, which is why it has never been done, except in limited, temporary trials. The biggest problem is the little word in the middle: basic. Without spending implausible amounts, the best that UBI can do is to make its recipients poor. That is much better than being dead, if that were the alternative, but it is nowhere near good enough. As the noted economist John Kay said, “either the basic income is impossibly low, or the expenditure on it is impossibly high.”

ABUNDANCE

Despite its failings, the proponents of UBI have hit upon a vital insight: in a post-jobs world, there are going to have to be substantial transfers of income and / or wealth from the minority who are still in paid employment, and from the owners of major assets. The problem is that if the cost of a good life remains high, then these transfers will be onerous, and therefore avoided.

